SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: ***Apply image enhancement and segmentation techniques for extracting meaningful regions from images.*** **(BTL 3)**

Assessment Tool Weightage: 10%

Code of Conduct:

Iam _A.Shiva Shankar Reddy/192425174(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

A.Shiva Shankar Reddy

Signature of the student:

Rubrics for Evaluation

Criteria	*Weightage*	*Level 4 – Exemplary*	*Level 3 – Proficient*	*Level 2 – Developing*	*Level 1 – Beginning*	*Marks(100)*
Understanding of Concepts	*25%*	*Demonstrates thorough, accurate understanding of concepts*	*Good understanding with minor gaps*	*Basic understanding with some misconceptions*	*Poor understanding; major misconceptions*	
Explanation	*25%*	*Detailed, well-explained answers with relevant examples*	*Clear explanation with adequate details*	*Limited explanation; lacks depth*	*Poor or unclear explanation*	
Application	*20%*	*Effectively applies concepts with strong analytical ability*	*Applies concepts with minor errors*	*Limited application with weak analysis*	*No application or incorrect analysis*	
Organization	*15%*	*Well-structured answers with logical flow and coherence*	*Organized with minor issues in flow*	*Basic structure, lacks clarity*	*Poor organization, incoherent answers*	
Accuracy	*10%*	*Completely accurate and covers all required points*	*Mostly accurate with minor omissions*	*Partially correct with missing points*	*Incorrect answers with major gaps*	
Clarity	*5%*	*Clear, neat, professional presentation*	*Understandable with minor issues*	*Limited clarity, readability issues*	*Poorly written, difficult to understand*	

1.An image contains high-frequency noise.Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

Answer:

The most suitable filter is the Low-Pass Filter (LPF), preferably a Gaussian Low-Pass Filter (GLPF).

Working:

- The image is transformed into the frequency domain using the Fourier Transform.
- High-frequency components mainly represent noise, sharp intensity variations, and fine details.
- The Low-Pass Filter allows low-frequency components to pass while attenuating high-frequency components.
- The filtered image is converted back to the spatial domain using the Inverse Fourier Transform (IFT).

Justification:

Since noise generally exists in the high-frequency region, suppressing these frequencies effectively reduces unwanted noise while preserving the overall image structure.

Impact on Image Quality:

- Reduces high-frequency noise.
- Produces a smoother image.
- Preserves the overall appearance.
- Slightly blurs edges because some edge information is also high-frequency

2.You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.

Answer:

The appropriate technique is the Gaussian Low-Pass Filter (GLPF).

Working:

- The image is transformed into the frequency domain using the FFT.
- The Gaussian filter applies a smooth weighting function that gradually decreases as the distance from the center frequency increases.
- Low-frequency components are retained, while high-frequency noise is gradually suppressed.
- The filtered image is reconstructed using the Inverse FFT.

Application:

GLPF is widely used in:

- Medical image preprocessing
- Satellite image enhancement
- Digital photography
- Image restoration

Advantages:

- Preserves important image information.
- Removes random noise effectively.
- Produces natural-looking smoothing.
- Avoids ringing artifacts common in Ideal LPF.

3.An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.

Answer:

The appropriate filter is the High-Pass Filter (HPF).

Working:

- The image is converted to the frequency domain.
- The HPF blocks low-frequency components and allows high-frequency components to pass.
- High-frequency components contain edge and fine-detail information.
- The enhanced image is reconstructed using the Inverse Fourier Transform.

Justification:

Edges represent rapid intensity changes, which correspond to high frequencies. Enhancing these frequencies makes object boundaries more prominent.

Effect on Frequency Components:

- Low frequencies are suppressed.
- High frequencies are amplified.
- Image becomes sharper.
- Fine details become more visible.

Applications:

- Medical imaging
- Object recognition
- Industrial inspection
- Remote sensing

4.A smooth version of an image is required.Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.

Answer:

The appropriate filter is the High-Pass Filter (HPF).

Working:

- The image is converted to the frequency domain.
- The HPF blocks low-frequency components and allows high-frequency components to pass.
- High-frequency components contain edge and fine-detail information.
- The enhanced image is reconstructed using the Inverse Fourier Transform.

Justification:

Edges represent rapid intensity changes, which correspond to high frequencies. Enhancing these frequencies makes object boundaries more prominent.

Effect on Frequency Components:

- Low frequencies are suppressed.
- High frequencies are amplified.
- Image becomes sharper.
- Fine details become more visible.

Applications:

- Medical imaging
- Object recognition
- Industrial inspection
- Remote sensing

5. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

Answer:

The appropriate segmentation technique is Threshold Segmentation (Global Thresholding).

Working:

- Select a threshold value (T).
- Compare every pixel intensity with T.
- Pixels with intensity greater than T are classified as the object.
- with intensity lower than T become the background.

Justification Pixels:

When there is a significant intensity difference between the object and background, thresholding provides accurate segmentation with minimal computation.

Effectiveness:

- Fast and simple.
- High accuracy for images with clear contrast.
- Suitable for document processing and industrial inspection.

6. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.

Answer:

The suitable method is Otsu's Thresholding.

Working:

- Analyze the histogram of image intensities.
- Automatically determine the threshold that minimizes intra-class variance.
- Pixels above the threshold become the foreground.
- Pixels below the threshold become the background.

Application:

- Medical imaging
- Optical Character Recognition (OCR)
- Industrial quality inspection

Advantages:

- Automatic threshold selection.
- No manual tuning required.
- Effective when the histogram is bimodal.
- Produces reliable foreground-background separation.

7. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

Answer:

The most suitable method is the Canny Edge Detection Algorithm.

Working:

1. Smooth the image using a Gaussian filter.
2. Compute intensity gradients.
3. Perform Non-Maximum Suppression.
4. Apply Double Thresholding.
5. Use Edge Tracking by Hysteresis to retain true edges.

Justification:

Canny detects strong edges while suppressing noise and minimizing false detections.

Effectiveness:

- High edge detection accuracy.
- Good localization.
- Continuous edges.
- Low noise sensitivity.

8. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.

Answer:

The suitable technique is Region Growing Segmentation.

Working:

- Select one or more seed pixels.
- Compare neighboring pixels with the seed.
- If intensity similarity satisfies a predefined criterion, include the pixel in the region.
- Continue until no more similar neighboring pixels remain.

Reasoning:

Adjacent pixels belonging to the same object usually have similar intensity values; therefore, they naturally form homogeneous regions.

Advantages:

- Produces connected regions.
- Preserves object boundaries.
- Effective for medical and satellite images.

9. Detected edges in an image appear broken. Explain and justify a method to improve edge continuity, including its role in post-processing.

Answer:

The appropriate method is Morphological Closing.

Working:

Morphological Closing consists of:

1. Dilation – expands object boundaries and connects broken edges.
2. Erosion – restores the original object size while keeping connected edges.

Role in Post-Processing:

- Connects discontinuous edge segments.
- Fills small gaps.
- Produces continuous object boundaries.
- Improves contour detection.

Applications:

- Medical imaging
- Object detection
- Shape analysis
- Industrial inspection

10. You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.

Answer:

The appropriate technique is Contour Detection.

Working:

1. Convert the image into a binary image using thresholding.
2. Detect object boundaries using contour tracing algorithms.
3. Store boundary points representing each object's outline.
4. Draw or analyze the detected contours.

Boundary Identification:

Contours identify pixels where the foreground meets the background, accurately outlining the object's shape.

Advantages:

- Provides accurate object boundaries.
- Enables shape analysis.
- Supports object recognition and measurement.
- Widely used in computer vision applications such as defect detection, medical imaging, and robotics.